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Supplementary Material S3: Detailed Assumption Analysis

Implicit Assumptions Revealed Through UML Reconstruction

Overview

This document provides detailed results from the quantitative assumption analysis conducted across three published *Aedes* population models. Two independent reviewers identified assumptions from both original publications and UML reconstructions.

Methodology

Reviewers

- **Reviewer A:** Ecological modeler, 8 years experience, no prior UML experience
- **Reviewer B:** Software engineer, 5 years experience in ecological modeling systems

Assumption Categories

1. **Structural assumptions:** Which components exist and how they relate
2. **Behavioral assumptions:** Process ordering, transition conditions, timing
3. **Boundary assumptions:** What is included/excluded from the model

Protocol

1. Reviewers independently read original publication
2. Listed all identifiable assumptions (30 min per model)
3. Received UML reconstruction (no additional explanation)
4. Listed all assumptions visible from UML (30 min per model)
5. Classifications compared and reconciled

Model 1: Lončarić & Hackenberger (2013)

Reference

Lončarić, Ž., & Hackenberger, B. K. (2013). Stage and age structured *Aedes vexans* and *Culex pipiens* (Diptera: Culicidae) climate-dependent matrix population model. *Theoretical Population Biology*, 83, 82-94. <https://doi.org/10.1016/j.tpb.2012.08.002>

Assumptions Identified from Original Publication

#	Assumption	Category	Source in Paper
1	Temperature-dependent development (degree-days)	Behavioral	Equations 3-5
2	Photoperiod-triggered diapause	Behavioral	Section 2.3
3	Density-dependent larval mortality	Structural	Equation 8

Total from original: 3

Additional Assumptions Revealed by UML

#	Assumption	Category	UML Element	Status in Original
4	Synchronous daily time step	Behavioral	Activity diagram: sequential process ordering	Implicit

#	Assumption	Category	UML Element	Status in Original
5	Instantaneous stage transitions	Behavioral	State machine: no transition duration states	Implicit (inferrable)
6	Homogeneous population (no individual variation)	Structural	Class diagram: no variance attributes	Implicit
7	Single breeding site type	Structural	Class diagram: one BreedingSite class	Implicit
8	Immediate oviposition after gonotrophic cycle	Behavioral	State machine: Gravid → HostSeeking direct	Implicit (inferrable)

Total from UML: 8 Newly revealed: 5

Reviewer Agreement

- Cohen's $\kappa = 0.82$ (almost perfect agreement)
- Disagreement on #5: Reviewer A marked as "explicit," Reviewer B as "implicit (inferrable)"
- Resolution: Classified as "implicit (inferrable)" - inferrable from equations but not stated

Model 2: Cailly et al. (2012)

Reference

Cailly, P., Tran, A., Balenghien, T., L'Ambert, G., Toty, C., & Ezanno, P. (2012). A climate-driven abundance model to assess mosquito control strategies. *Ecological Modelling*, 227, 7-17.

Assumptions Identified from Original Publication

#	Assumption	Category	Source in Paper
1	Temperature-dependent vital rates (Briere function)	Behavioral	Section 2.2
2	Rainfall affects breeding site availability	Structural	Section 2.3
3	Nulliparous/parous adult distinction	Structural	Figure 1
4	Control measures as mortality modifiers	Structural	Section 2.5

Total from original: 4

Additional Assumptions Revealed by UML

#	Assumption	Category	UML Element	Status in Original
5	Multiple breeding site types with different dynamics	Structural	Class diagram: SiteType enum	Implicit
6	Sex ratio determination at emergence	Behavioral	State machine: determineSex() at Pupa→Adult	Implicit
7	Bloodmeal required before each oviposition	Behavioral	State machine: HostSeeking→Bloodfed→Gravid cycle	Implicit (inferrable)
8	No adult immigration/emigration	Boundary	Class diagram: closed population	Implicit
9	Environmental drivers affect all stages uniformly	Structural	Sequence diagram: single Temperature object	Partially explicit
10	Deterministic dynamics (no stochasticity)	Behavioral	No <> stereotypes	Implicit
11	No predation mortality component	Boundary	Class diagram: no Predator class	Implicit

Total from UML: 11 Newly revealed: 7

Reviewer Agreement

- Cohen's $\kappa = 0.78$ (substantial agreement)
- Disagreement on #9: Debate whether "partially explicit" counts as "revealed"
- Resolution: Included as revealed since full scope became apparent only through UML

Model 3: Tran et al. (2013)

Reference

Tran, A., L'Ambert, G., Lacour, G., Benoît, R., Demarchi, M., Cros, M., Cailly, P., Aubry-Kientz, M., Balenghien, T., & Ezanno, P. (2013). A rainfall- and temperature-driven abundance model for *Aedes albopictus* populations. *International Journal of Environmental Research and Public Health*, 10(5), 1698–1719.

Assumptions Identified from Original Publication

#	Assumption	Category	Source in Paper
1	Rainfall-dependent carrying capacity	Structural	Equation 4
2	Temperature-dependent development	Behavioral	Section 2.2

Total from original: 2

Additional Assumptions Revealed by UML

#	Assumption	Category	UML Element	Status in Original
3	Four larval instars with separate thresholds	Structural	State machine: L1→L2→L3→L4 states	Implicit
4	Pupal stage non-feeding	Behavioral	Class diagram: no feedingRate in Pupa	Implicit
5	Adult males do not affect population dynamics	Structural	Class diagram: reproduction in Female only	Implicit
6	Single gonotrophic cycle length (no variation)	Behavioral	Class diagram: fixed gonotrophic-CycleLength	Implicit
7	No egg bank (all eggs hatch or die)	Boundary	State machine: no persistent egg state	Implicit
8	Habitat desiccation causes egg mortality	Behavioral	Activity diagram: desiccation mortality branch	Implicit
9	Process order: development before mortality	Behavioral	Activity diagram: explicit ordering	Implicit

Total from UML: 9 Newly revealed: 7

Reviewer Agreement

- Cohen's $\kappa = 0.85$ (almost perfect agreement)
- Minor disagreement on #4: Both agreed it was implicit but debated importance
- Resolution: Included all assumptions regardless of perceived importance

Summary Statistics

Assumptions by Model

Model	Original	After UML	Newly Revealed	% Increase
Lončarić & Hackenberger (2013)	3	8	5	167%
Cailly et al. (2012)	4	11	7	175%
Tran et al. (2013)	2	9	7	350%
Mean	3.0	9.3	6.3	210%

Assumptions by Category

Category	Original Total	UML Total	Newly Revealed
Structural	4	13	9
Behavioral	4	12	8
Boundary	1	3	2
Total	9	28	19

Inter-Rater Reliability

Model	Cohen's κ	Interpretation
Lončarić & Hackenberger (2013)	0.82	Almost perfect
Cailly et al. (2012)	0.78	Substantial
Tran et al. (2013)	0.85	Almost perfect
Mean	0.82	Almost perfect

Interpretation scale (Landis & Koch, 1977): - 0.81–1.00: Almost perfect - 0.61–0.80: Substantial - 0.41–0.60: Moderate - 0.21–0.40: Fair - 0.00–0.20: Slight

Categories of Revealed Assumptions

1. Temporal Structure (6 assumptions)

- Process ordering within time steps
- Synchronous vs. asynchronous updates
- Instantaneous vs. duration-based transitions

2. Spatial Homogeneity (4 assumptions)

- Single vs. multiple breeding site types
- Closed population boundaries
- Uniform environmental effects

3. Population Structure (5 assumptions)

- Individual variation (or lack thereof)
- Sex-specific dynamics
- Age/stage sub-structure

4. Process Boundaries (4 assumptions)

- Included vs. excluded mortality sources
 - Immigration/emigration
 - Predation effects
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Implications

1. **Narrative descriptions consistently underspecify models** - On average, only 32% of assumptions were identifiable from original publications.
 2. **UML reveals behavioral assumptions most effectively** - Process ordering and transition logic became explicit through activity and state machine diagrams.
 3. **High inter-rater agreement validates UML interpretability** - Reviewers from different backgrounds reached similar conclusions from UML specifications.
 4. **Boundary assumptions are frequently implicit** - What is *excluded* from models is rarely stated explicitly.
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References

Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, 33(1), 159-174.

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